PLANT GROWTH SYSTEM
ANNEX I InventoryBridge

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Project Title: Plant Growth System	**Document:** ANNEX I InventoryBridge

Inventory Bridge — Setup Guide

1. Add the Inventory Bridge

Attach the **Inventory Bridge** component to your **Plant Entity**.

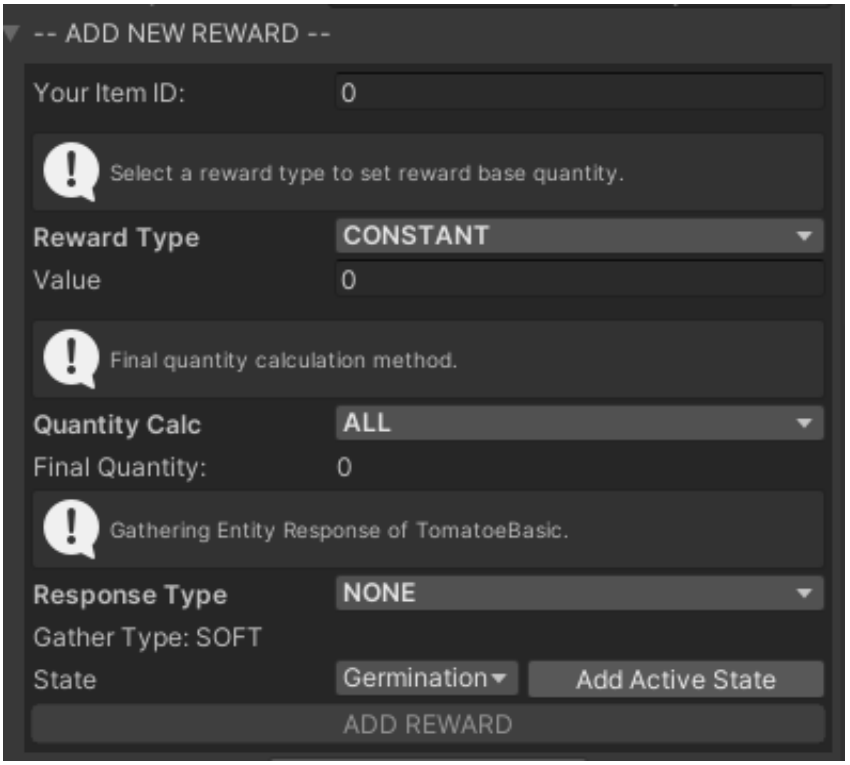
This component is responsible for handling all reward logic and connecting your plant with any external inventory system.

2. What is a Reward?

A **Reward** defines what the plant will generate when interacted with.

Each reward can:

- Grant an item (via Item ID)
- Be limited to specific plant states
- Calculate quantity dynamically
- Trigger additional effects (like modifying variables or removing the plant)



3. Configuring a Reward

Each reward contains several configurable fields:

🔹 Item ID

Defines which item will be generated.
This ID must match your **external inventory system.**

🔹 Active States

List of plant states where this reward is valid.
If the plant is not in one of these states, the reward will not trigger.

🔹 Reward Type

Determines how the quantity is calculated:

- **CONSTANT** → Uses a fixed value

- **REFERENCE** ➜ Uses external variables
- **DIFFERENCE** ➜ Calculates based on the difference between values

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🔹 Quantity Calculation

Defines how the final amount is applied:

- **ALL** ➜ Uses the full calculated value
- **RANDOM** ➜ Random value within range (MIN & MAX)
- **RELATIVE** ➜ Uses two values:
    - The first acts as an **offset**, allowing you to add a minimum or subtract from the final reward
    - The second defines a **percentage factor** applied to the final calculated amount

🔹 Response Type

Defines what happens after the reward is applied:

- **NONE** ➜ No additional effect (SOFT)
- **ADD VARIABLE** ➜ Adds value to a variable (SOFT)
- **SUBTRACT VARIABLE** ➜ Subtracts value (SOFT)
- **REMOVE PLANT** ➜ Removes the plant after harvesting (HARD)

4. Gather Types

Rewards are divided into two categories:

- **Soft Gather**
  - Used for non-destructive interactions
  - The plant remains after collecting rewards
  - Typically used for repeated harvesting (e.g. fruits, resources over time)

Call:

public List <PGSRewardsResults> = inventoryBridge.EvaluateRewardsSoft(plant);

You can check if a plant in a determined state has a Soft Reward.

public bool CheckGatherSoft()

- **Hard Gather**
  - Used for destructive interactions
  - The plant may be removed after collecting rewards
  - Typically used for full harvest (e.g. cutting the plant)

Call:

public List <PGSRewardsResults> = inventoryBridge.EvaluateRewardsHard(plant);

You can check if a plant in a determined state has a Hard Reward.

public bool CheckGatherHard()

Returned Data

Both methods return a list of `PGSRewardResult`:

public PGSRewardResult(int itemID, int quantity)

Each result contains:

- **itemID → The ID of the item to add to your inventory**
- **quantity → The amount generated**

5. Viewing Rewards

The system provides two ways to visualize rewards inside the editor:

- **By State**

Rewards are grouped by plant state.
This allows you to quickly see what each state will produce.

- **By Reward**

Displays each reward independently, regardless of state. Useful for debugging and fine-tuning individual rewards.

Community & Shared Content

This plugin is designed to grow alongside its community.

The long-term goal is to provide:

- A shared Plant Prefab database
- A collection of reusable Soils, Plants, and configurations
- Community-driven improvements and discussions

All prefabs will be fully compatible with the simulation system and can be downloaded and used directly in your projects.

Discord Server

https://discord.gg/DPWHUP7vWA

A dedicated Discord server is available for:

- Questions and troubleshooting
- Suggestions and feature requests
- Sharing Plant and Soil Prefabs
- Discussing simulation setups and design ideas

Community feedback will directly influence future updates of the plugin.